Probability

Year 4 Pt 2 · **Mark Scheme**

Lesson Worksheet — Answers

Probability Basics

1. 1/6

2. 1/2 (odd numbers 1, 3, 5, 7 out of 8 sections)

3. E) 1

4. 0.6

5. impossible, unlikely, evens, likely, certain

6. 50%

7. C) Evens

8. 1 (it is certain)

9. Heads, Tails

10. 3/10

11. A, C and E (0, 0.5 and 1 are valid probabilities; 1.2 and −0.3 are not, as probabilities must lie between 0 and 1 inclusive)

12. B) 1/3 (rolling a 5 or 6 is 2 out of 6 outcomes)

13. 15%

14. 1, 2, 3, 4, 5, 6

15. 3/4 (P(red) = 1/4, so P(not red) = 1 − 1/4 = 3/4)

Applying Probability

16. 5/10 = 1/2

17. 0.6

18. D) 0.6

19. HH, HT, TH, TT; P(one head and one tail) = 2/4 = 1/2

20. 60%

21. C) 3/4

22. 1/3 (3 of the remaining 9 counters are red)

23. A, B, C and D are true (E is false — among 1–6 only the number 5 is a multiple of 5, so P = 1/6, not 1/2)

24. 3/5

25. 3/10

26. 28 blue counters (P(red) = 0.3, so red = 0.3 × 40 = 12; blue = 40 − 12 = 28)

27. A) 1/25

28. 1/3 (6/10 × 5/9 = 30/90 = 1/3)

29. 0.025

30. A, B, D and E are true (C is false — P(green) = 3/10 but P(blue) = 4/10, so they are not equal)

Probability Challenge

31. 10 green counters (red = 1/3 × 24 = 8; blue = 1/4 × 24 = 6; green = 24 − 8 − 6 = 10)

32. C) 1/6 (6 of the 36 equally likely outcomes give a total of 7)

33. 3/28 (first counter blue: 3/8; second counter blue: 2/7; 3/8 × 2/7 = 6/56 = 3/28)

34. 1/4 (P(even) = 1/2 and P(heads) = 1/2; these are independent, so 1/2 × 1/2 = 1/4)

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35. A, B, C and D are true (E is false — 6 of the 12 numbers are even, so P = 1/2, not 1/3)

36. A) 4/15 (4/10 × 6/9 = 24/90 = 4/15)

37. 1/25 (1/5 × 1/5)

38. 15/36 = 5/12 (of the 36 outcomes, 6 have equal scores; the remaining 30 split evenly between 'second higher' and 'first higher', giving 15 each)

39. 12/19 (originally 8 red and 12 green out of 20; after removing one red, 7 red and 12 green remain out of 19 counters, so P(green) = 12/19)

40. A, B, C and D are true (E is false — P(no heads) = P(TT) = 1/4, not 1/2)

41. 16/25 (the product is odd only if both spins are odd: P(both odd) = 3/5 × 3/5 = 9/25; so P(even product) = 1 − 9/25 = 16/25)

42. C) 11/15 (P(same colour) = P(both red) + P(both blue) = 1/5 + 1/15 = 4/15, so P(different colours) = 1 − 4/15 = 11/15)

43. 0.3 (0.6 × 0.5)

44. 2/3 (1 − 1/3)

45. 1/3 (of the 9 equally likely outcome pairs, 3 give a total greater than 4: (2,3), (3,2) and (3,3))

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Homework Worksheet — Answers

Probability Basics

1. 1/6

2. 2/5 (even numbers 2 and 4 out of 1, 2, 3, 4, 5)

3. A) 0

4. 0.25

5. impossible, unlikely, evens, likely, certain

6. 50%

7. D) Likely

8. 0 (it is impossible)

9. 1, 2, 3

10. 1/10

11. A, C and E (1, 0.75 and 0 are valid probabilities; −0.5 and 2 are not)

12. B) 1/3 (rolling a 1 or 2 is 2 out of 6 outcomes)

13. 85%

14. HH, HT, TH, TT

15. 2/5 (2 red sections out of 5)

Applying Probability

16. 4/10 = 2/5

17. 0.375

18. C) 0.45

19. HH, HT, TH, TT; P(at least one tail) = 3/4

20. 60%

21. B) 1/2

22. 2/7 (2 of the remaining 7 counters are purple)

23. A, B, C and D are true (E is false — only 5 is a multiple of 5 among 1–8, so P = 1/8, not 1/4)

24. 2/5

25. 1/5

26. 11 yellow (P(green) = 0.45, so green = 0.45 × 20 = 9; yellow = 20 − 9 = 11)

27. A) 1/16

28. 2/9 (5/10 × 4/9 = 20/90 = 2/9)

29. 0.02

30. A, B, C and D are true (E is false — red and green together make 5 of the 10 counters, so P = 0.5, not 0.6)

Standard / Advanced

31. 12 red counters (P(red) = 3/9 = 1/3; 1/3 × 36 = 12)

32. C) 8 (2 × 2 × 2)

33. 1/2 (factors of 12 that are 10 or less: 1, 2, 3, 4, 6 — that is 5 of the 10 sections, so P = 5/10 = 1/2)

34. 50% (P(blue) = 1/4 = 25%; doubled = 50%)

35. 7, with probability 6/36 = 1/6

36. 5/6 (red is 1 part of 6, so P(red) = 1/6 and P(not red) = 5/6)

37. D) 7/22 (7/12 × 6/11 = 42/132 = 7/22)

38. 0.1 (0.25 × 0.4)

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39. A) 28 (P(blue) = 1 − 2/7 = 5/7; total = 20 ÷ 5/7 = 28)

40. 0.2 (0.6 × 1/3)

41. 1/2 (an odd total needs one odd and one even score; P = 2 × (1/2 × 1/2) = 1/2)

42. A, B, C and D are true (E is false — P(both red) = 1/3 but P(both blue) = 2/15, so they are not equal)

43. 1/6 (only the pairs (4,6), (5,5), (5,6), (6,4), (6,5) and (6,6) give a product over 20 — 6 of the 36 outcomes)

44. 9 red (red:blue = 3:1, so red = 3/4 × 12 = 9)

45. 2/9 (there are C(5,2) = 10 ways to choose two even tickets out of C(10,2) = 45 total ways; 10/45 = 2/9)

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